Problem 11

Evaluate the sums for the following series:

$$1^\circ \quad S_1 = \sum_{n=1}^{\infty} q^n \sin(nx) \quad (|q| < 1); \qquad 2^\circ \quad S_2 = \sum_{n=1}^{\infty} q^n \cos(nx) \quad (|q| < 1).$$

Solution

Approach: Complex Exponential Method

We will solve both series simultaneously using Euler's formula. Define the complex series:

$$Z = \sum_{n=1}^{\infty} q^n e^{inx} = \sum_{n=1}^{\infty} q^n (\cos(nx) + i \sin(nx)).$$

By linearity of summation:

$$Z = \sum_{n=1}^{\infty} q^n \cos(nx) + i \sum_{n=1}^{\infty} q^n \sin(nx) = S_2 + iS_1.$$

Therefore, once we find Z, we can extract:

$$S_1 = \mathrm{Im}(Z) \quad \text{and} \quad S_2 = \mathrm{Re}(Z).$$

Step 1: Evaluating the Complex Geometric Series

The series Z can be rewritten as:

$$Z = \sum_{n=1}^{\infty} (qe^{ix})^n.$$

This is a geometric series with first term $a = qe^{ix}$ and common ratio $r = qe^{ix}$. Since $|r| = |q||e^{ix}| = |q| < 1$, the series converges to:

$$Z = \frac{qe^{ix}}{1 - qe^{ix}}.$$

Step 2: Simplifying the Complex Expression

We need to express Z in the form $A + iB$ where A and B are real. First, write $e^{ix} = \cos x + i \sin x$:

$$Z = \frac{q(\cos x + i \sin x)}{1 - q(\cos x + i \sin x)} = \frac{q \cos x + iq \sin x}{(1 - q \cos x) - iq \sin x}.$$

To separate real and imaginary parts, multiply numerator and denominator by the complex conjugate of the denominator:

$$Z = \frac{(q \cos x + iq \sin x)[(1 - q \cos x) + iq \sin x]}{[(1 - q \cos x) - iq \sin x][(1 - q \cos x) + iq \sin x]}.$$

The denominator becomes:

$$(1 - q \cos x)^2 + (q \sin x)^2 = 1 - 2q \cos x + q^2 \cos^2 x + q^2 \sin^2 x = 1 - 2q \cos x + q^2.$$

The numerator is:

$$\begin{aligned}
&(q \cos x + iq \sin x)[(1 - q \cos x) + iq \sin x] \\
&= q \cos x(1 - q \cos x) + iq \cos x \cdot q \sin x + iq \sin x(1 - q \cos x) + i^2 q \sin x \cdot q \sin x \\
&= q \cos x - q^2 \cos^2 x + iq^2 \cos x \sin x + iq \sin x - iq^2 \sin x \cos x - q^2 \sin^2 x \\
&= (q \cos x - q^2 \cos^2 x - q^2 \sin^2 x) + i(q \sin x) \\
&= q \cos x - q^2 + iq \sin x.
\end{aligned}$$

Therefore:

$$Z = \frac{q\cos x - q^2 + iq\sin x}{1 - 2q\cos x + q^2}.$$

Step 3: Extracting Real and Imaginary Parts

From the expression above:

$$\mathrm{Re}(Z) = \frac{q\cos x - q^2}{1 - 2q\cos x + q^2} = S_2,$$

$$\mathrm{Im}(Z) = \frac{q\sin x}{1 - 2q\cos x + q^2} = S_1.$$

Final Answers

Series 1°: $\sum_{n=1}^{\infty} q^n \sin(nx)$

$$\boxed{S_1 = \sum_{n=1}^{\infty} q^n \sin(nx) = \frac{q\sin x}{1 - 2q\cos x + q^2}, \quad |q| < 1.}$$

Series 2°: $\sum_{n=1}^{\infty} q^n \cos(nx)$

$$\boxed{S_2 = \sum_{n=1}^{\infty} q^n \cos(nx) = \frac{q\cos x - q^2}{1 - 2q\cos x + q^2}, \quad |q| < 1.}$$

Verification

Note that the denominator can be factored as:

$$1 - 2q\cos x + q^2 = (1 - qe^{ix})(1 - qe^{-ix}) = |1 - qe^{ix}|^2 > 0$$

for $|q| < 1$, ensuring the formulas are well-defined.